THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

(KATAVI, MBEYA, NJOMBE, RUKWA, RUVUMA AND SONGWE)

FORM FOUR ZONAL MOCK II EXAMINATION

042 ADDITIONAL MATHEMATICS

TIME: 3:00 HOURS YEAR: 2025

INSTRUCTIONS

1. This paper consists of section A and B with a total of **fourteen (14)** questions.
2. Answer **all** questions.
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3. Section A carry **sixty (60)** marks and section B carry **forty (40)** marks.
4. NECTA Mathematical table and Non-programmable calculator may be used.
5. All communication devices and any unauthorized materials are **not** allowed in the examination room.
6. Write your **Examination Number** on every page of your answer sheets provided.
7. The following constant may be used.
   - Radius of the earth, R = 6370km
   - Pie, π = 3.14

SECTION A (60 Marks)

Answer **all** questions in this section.

1. (a) The expression $\frac{y}{a} - 4$ varies directly as the square of $\frac{1}{ax}$ if the constant of variation is 108 when $x = 3$, and $y = 16$. Find the value of a.

   (b) Thirty tractors can plough a piece of land for exactly twenty days. How many tractors of similar ability are required if there are only fifteen working days?

2. (a) The following frequency table represents the age of 160 people who visit a certain public library.

Age(years)	8 - 15	16 - 23	24 - 31	32 - 39	40 - 47	48 – 55	56 - 63
Frequency	12	29	40	44	20	12	3

   Calculate:
   (i) The median age
   (ii) The mode.

   (b) Songombingo is among students who didn't attend the lesson on computations of variance for given information, so is lacking variances computations skills. You as an expert student show how to compute the variance of the data 6, 7, 10, 13, 17, 19, 20, 25, and 27.

3. (a) Calculate the shortest distance of a point A (6,5) from the line $y = \frac{-5}{12}x - \frac{7}{6}$

   (b) Find the equation of the circle in expanded form given the center (2,3) and radius 5.

   (c) Given that the angle between two lines $3x - y + 7 = 0$ and $x - ny + 8 = 0$ is 60°, find the possible value(s) of n.

4. (a) Find the locus of Point P whose distance from the point $A(-1, 2)$ is twice its distance from the origin.

   (b) Calculate the possible values of the m, such that points $A(2m, 5)$, $B(2,1)$ and $C(m, 0)$ are collinear.

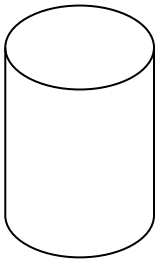
5. (a) Given that the equation $(3a + 1)x^2 + (a + 2)x + 1 = 0$ has equal roots, find the value(s) of *a.*

   (b) A piece of wire 56cm long is bent to form a rectangle with an area of $171cm^2$. Find the dimensions of the rectangle.

6. (a) Fill the blanks in the following table.

Name of a figure/ letter	Rotational symmetry	Number of lines of symmetry
……..	4	4
H	2	……
……..	Infinity	infinity

(b) Draw the plan and elevation of a cylinder drawn below.



7. (a) Prove that $\frac{\sin 2\theta}{1+\cos\theta} = \tan\theta$

(b) Solve the equation $1+ \cos\theta = 2\sin^2\theta$, for values of θ between 0^0 and 360^0.

8. (a) List down the next two numbers in the series $\frac{1}{1\times2} + \frac{1}{2\times6} + \frac{1}{4\times18} + \ldots\ldots\ldots$

(b) (i) Explain the divisibility rule of a number
(ii) Use the divisibility rule to show whether or not 24679 is divisible by seven

9. (a) Show that the statement $p \rightarrow (p \vee q)$ is tautology.

(b) Draw an electric network corresponding to the compound statement $R \wedge (P \vee Q \vee S)$.

(c) Determine the truth value of the proposition:
If $\sqrt[4]{625} - \sqrt[3]{64} \neq 1$ then $5^4 - 4^3 = 1$ and $18 \times 4 - 18 \times 9 = -18(9+4)$.

10. (a) By using set properties, simplify the expression.
(i) $(\mathrm{A} \cup \mathrm{B})' \cap A$
(ii) $(A - B) \cap (B - A)$

(b) In a class of twenty pupils, there are 12 who study Chemistry but not Biology, 4 who study Biology but not Chemistry and 1 who study neither Chemistry nor Biology.

How many pupils' study.

(i) Chemistry and Biology
(ii) Chemistry or Biology

SECTION B (40 marks)

Answer **all** questions in this section.

11.(a) If $x^2 - 1$ is a factor of the expression $x^3 - px^2 - qx - 2$ find;

i. The values of p and q

ii. The quotient when is divided by $x + 1$.

(b) The rational function is given as: $f(x) = \frac{x^2+2}{x^2-3x-4}$

i. Determine the vertical asymptote and horizontal asymptote.

ii. Find the intercepts

iii. Sketch the graph

12.(a) Differentiate $(2x - 1)^3$ with respect to x.

(b) Evaluate $\int_{-1}^{3}(4x^2 - 3x + 5)dx$

(c) Find the area of the region bounded by the curve $y = 2 + x - x^2$, x-axis, $x = -1$ and $x = 1$.

(d) Determine the volume of the solid figure formed after complete revolution of the curve $x^2 + y^2 = 4$ about the $y - axis$.

13.(a) If $P(A) = 0.4$, $P(B) = 0.7$ and $P(A \cap B) = 0.3$;
Find: (i) $P(A' \cap B)$ (ii)$P(A' \cap B')$

(b) How many different selections of 5 skirts can be made from 10 skirts?

(c) By factorization evaluate: $9! \div 6(7!)$

14.(a) Given the matrices $M = \begin{bmatrix} 2 & 4 \\ -1 & 3 \\ 3 & -2 \end{bmatrix}$ and $N = \begin{bmatrix} 1 & -2 \\ 3 & 4 \\ 5 & 2 \end{bmatrix}$, Find $M^T N$.

(b) Given the vectors $\boldsymbol{a} = 2\boldsymbol{i} + \boldsymbol{j} + \boldsymbol{k}$ and $\boldsymbol{b} = \boldsymbol{i} - \boldsymbol{j} + 2\boldsymbol{k}$
Find (i) $\boldsymbol{a} \times \boldsymbol{b}$
(ii) $\boldsymbol{a}.\boldsymbol{b}$

(c) Reflect the circle centered at P (-2,1) having radius 5 units through the line $x = 0$. Hence deduce its equation.